

We show that
neural
networks can
outperform
the numerical
simulator that
produced their
training data.

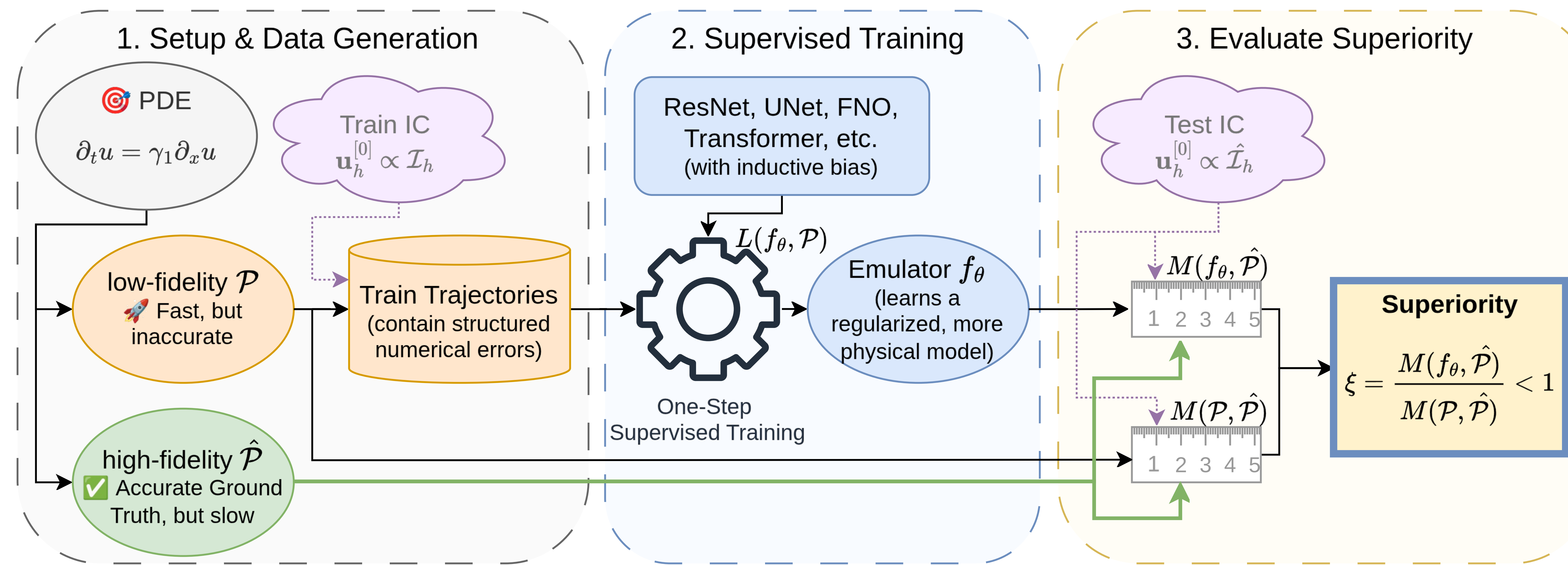


Project Page

Emulator Superiority

When Machine Learning for PDEs Surpasses its Training Data

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Differences between Training & Testing

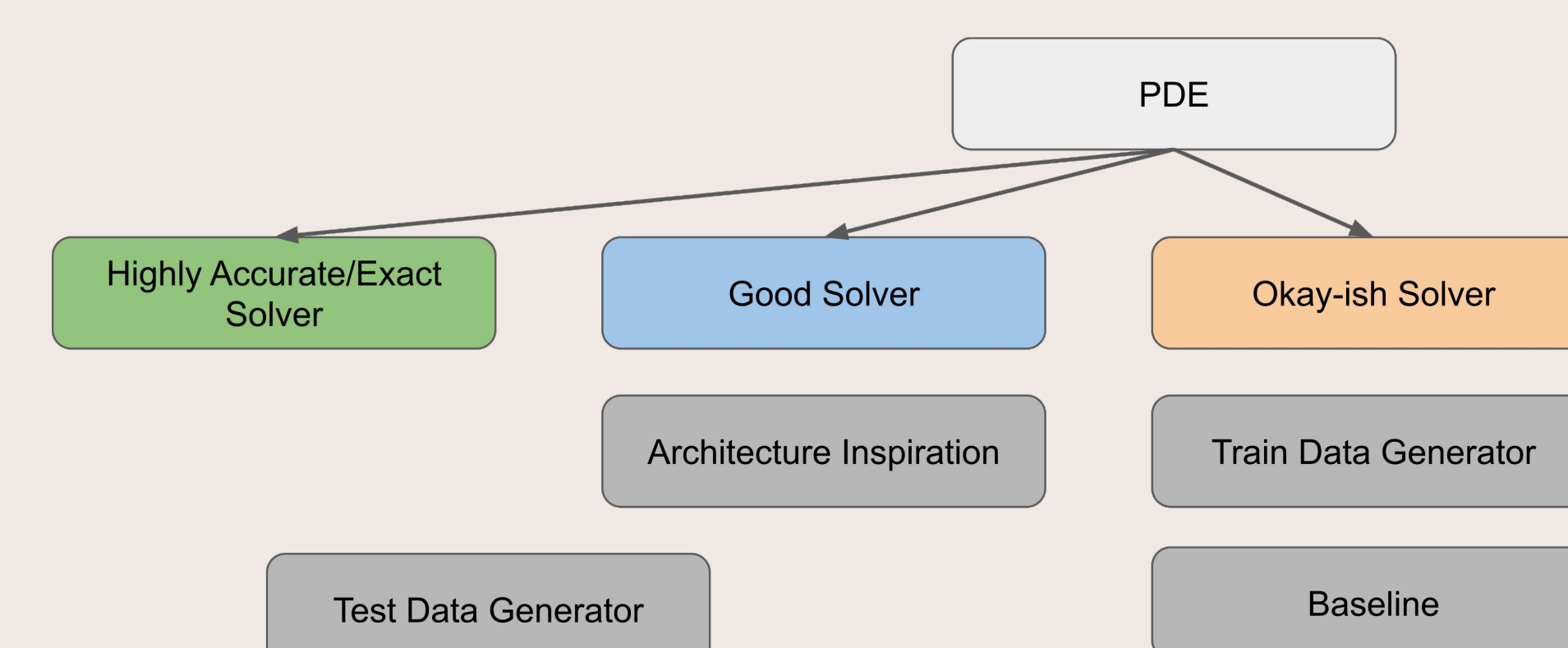
The training objective of autoregressive emulators (**one-step training**)

$$L(\theta) = \mathbb{E}_{\{\mathbf{u}_h^{[t]}, \mathbf{u}_h^{[t+1]}\} \sim \mathcal{T}_{\mathcal{P}_h}(\mathbf{u}_h^{[0]}, \mathbf{u}_h^{[0]} \sim \mathcal{I}_h} \left[\zeta \left(f_\theta(\mathbf{u}_h^{[t]}, \mathbf{u}_h^{[t+1]}) \right) \right]$$

differs from the inference goal (**multi-step prediction** often with different initial condition (IC) distribution).

$$e^{[t]} = \mathbb{E}_{\mathbf{u}_h^{[0]} \sim \hat{\mathcal{I}}_h} \left[\zeta \left(f_\theta^t(\mathbf{u}_h^{[0]}, \tilde{\mathcal{P}}_h^t(\mathbf{u}_h^{[0]})) \right) \right]$$

There are multiple ways to solve a given PDE.

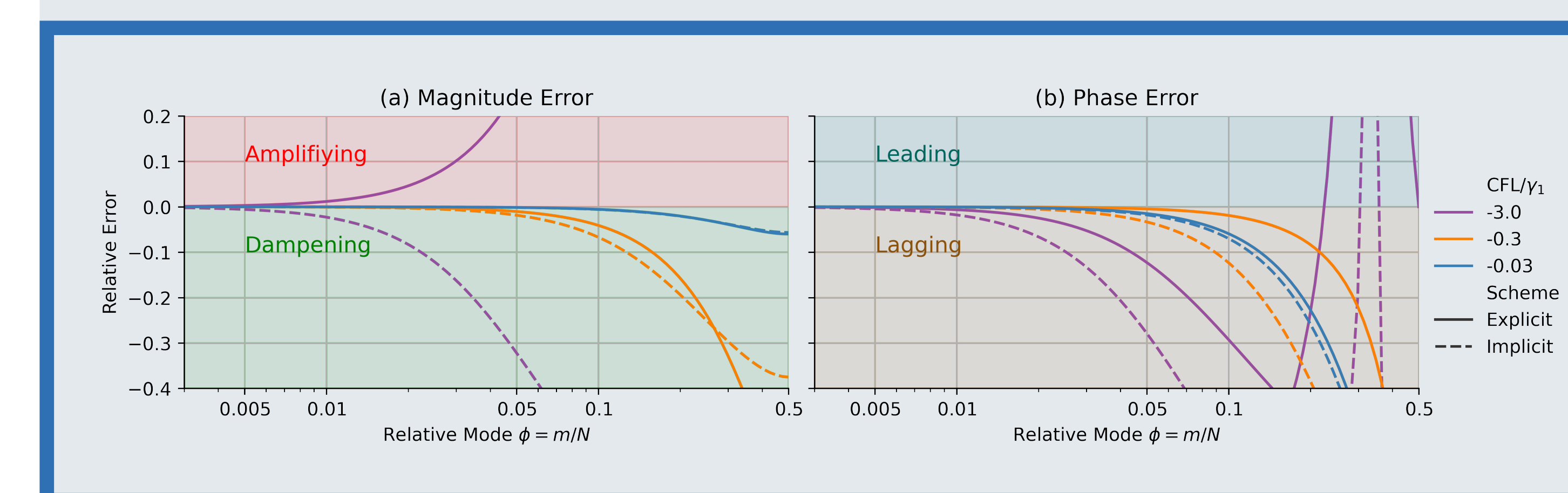


If trained on low-fidelity data, the emulator faithfully learns the one-step object. When choosing a high-fidelity test generator, it can become better than its low-fidelity training source and we observe **superiority** $\xi < 1$.

$$\xi = \frac{\text{Error}(\text{Emulator vs. } \checkmark \text{ Ground Truth})}{\text{Error}(\text{Training Data vs. } \checkmark \text{ Ground Truth})}$$

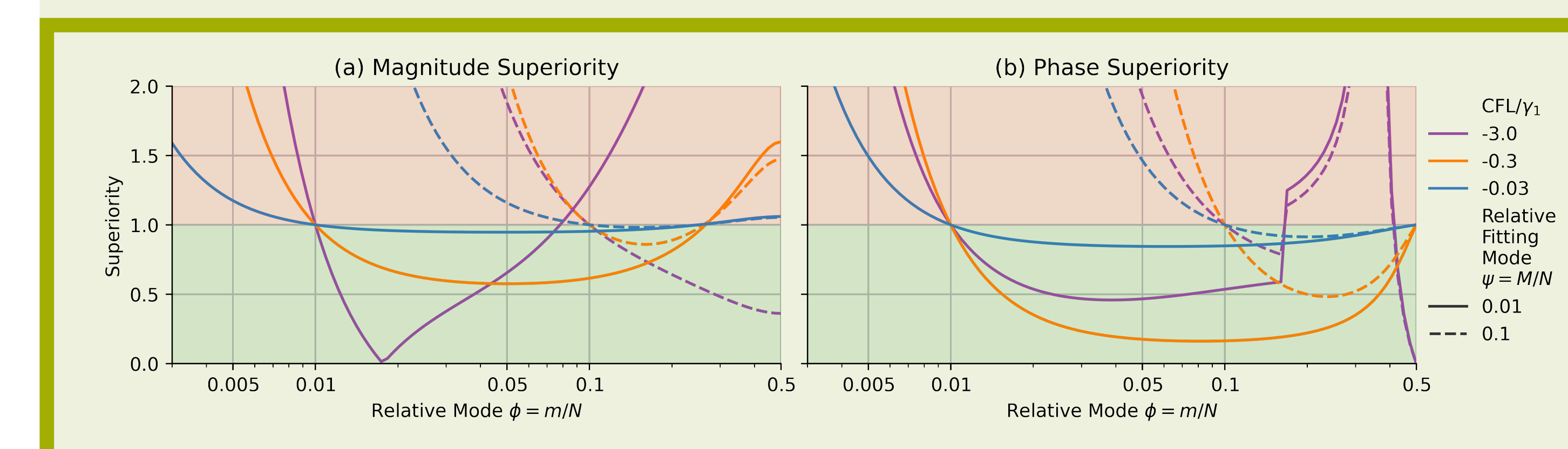
Numerical Solvers are Imperfect

Explicit & Implicit Scheme for Advection against (exact) **spectral solver**.



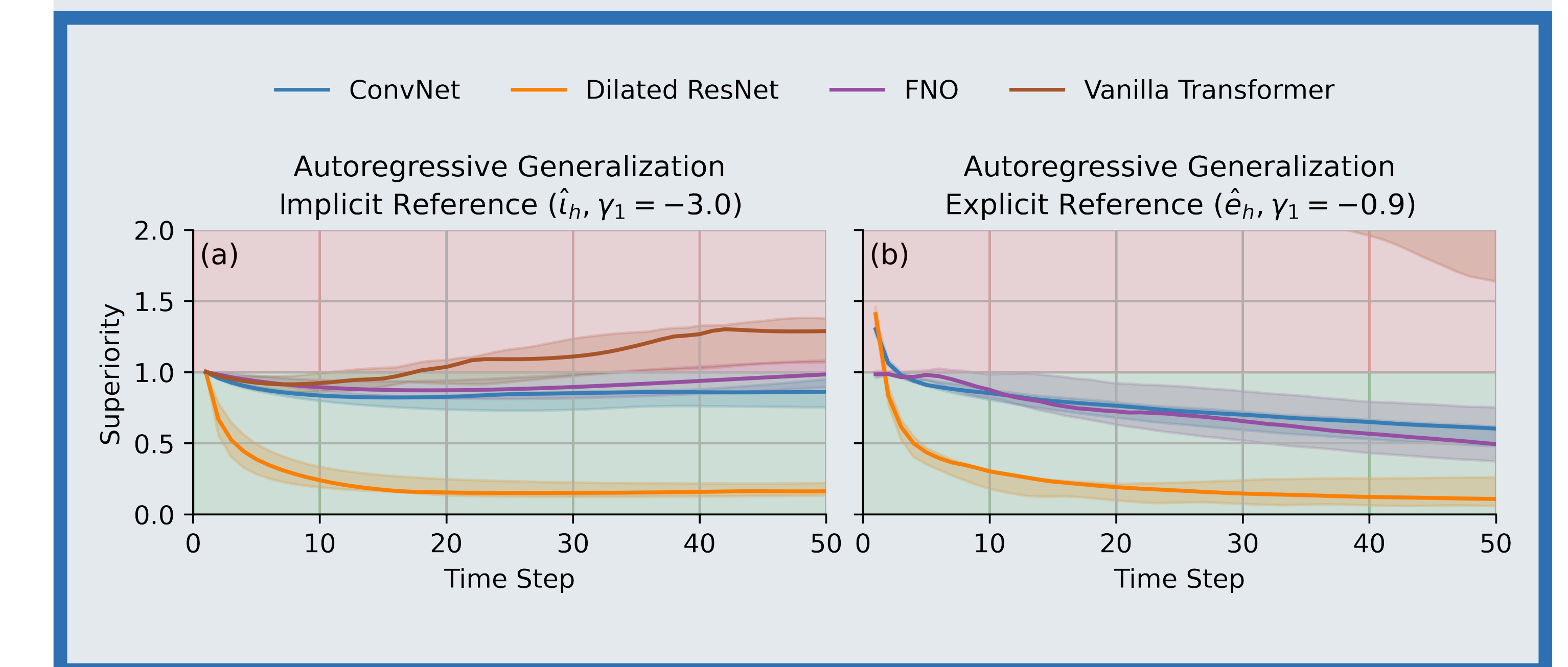
Superiority is Theoretically Provable

Linear Emulator for Advection Equation at CFL γ_1 trained on data of an implicit FOU scheme at relative mode $\psi = M/N$ and tested at $\phi = m/N$.



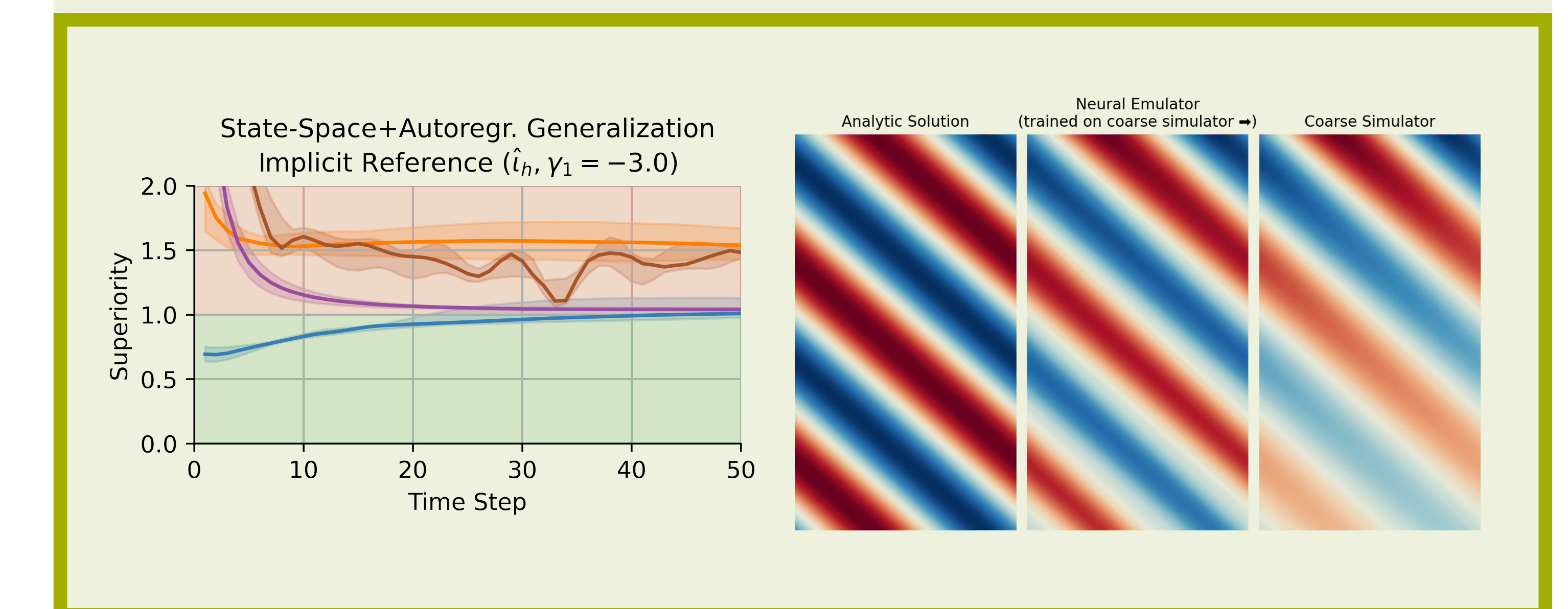
Nonlinear Emulators -> Autoregressive Sup.

- Nonlinear networks can achieve better test rollout performance than their training simulator, even when the initial condition distribution is identical -- due to superior error accumulation.
- Also holds against the more accurate explicit numerical scheme.



Autoregressive + State-Space Superiority

- Superiority in Advection noticeable by less numerical diffusion.
- Only the strongest inductive bias (ConvNet) achieves both state-space and autoregressive superiority.



Superior Burgers Shock Prediction

- Truncating (Non-)Linear Solver Iteration is common to reduce cost:
 - PISO Schemes for Navier-Stokes with Picard-1 style convection.
 - Position-Based Dynamics with partially resolved constraints.
- Leads to incorrect nonlinear behavior (here incorrect shock angle).
- Network trained on coarse solver (specifically on smooth initial frames) learns superior shock prediction (forward state & autoregr. superiority).

